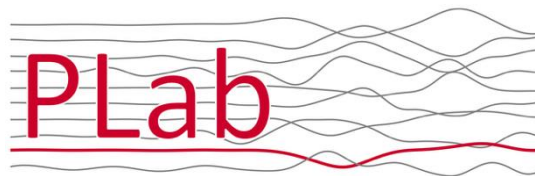


# Generational shift in Maga Rukai vowel alternations as a window into learning biases

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# Windows into phonological learning biases

Speakers, given sufficient exposure to an (unnatural) phonological pattern, can generally learn it well

**language change** as a natural laboratory for learnability of phonological patterns (Kiparsky 1968)

**heritage speakers** have less sustained input to their heritage L1, and may therefore be more sensitive to learning biases.

# Questions/goals of this talk

Empirical focus:  $\emptyset \sim V$  alternations in Maga Rukai

**Question:** how does this alternation change across two generations of speakers?

# Roadmap

Background, traditional account of  
Maga vowel alternations

**Q: why is it hard to learn?**



Comparing two generations of  
speakers, with different levels of  
Maga proficiency



**OT formalization:** can constraint re-  
ranking capture the difference  
between two generations? How can  
this ranking help us understand  
learning biases in phonology?

Data and description from  
Hsin (2000)

# Language details – Maga Rukai

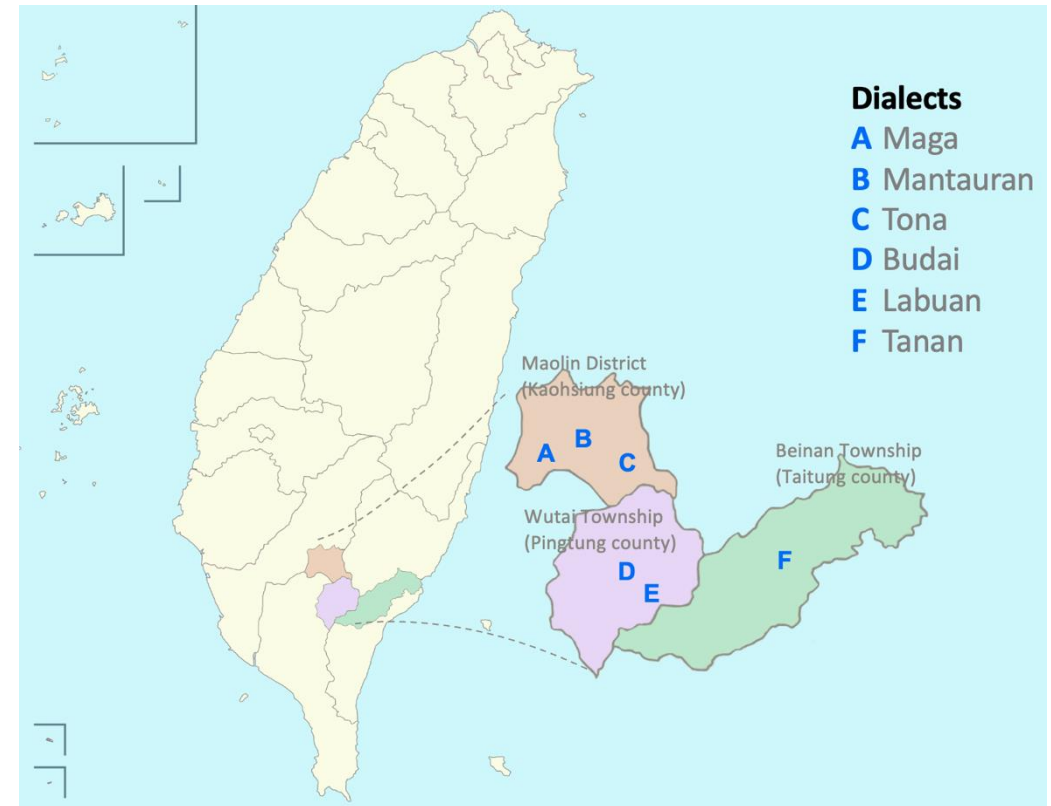
Austronesian language spoken in Southern Taiwan

Dialects: **Maga**, Tanan, Budai, Tona, Labuan, Mantauran

Low mutual intelligibility

Maolin Township

- population ~866
- Much fewer speakers



# A little bit on negation

| Type         | Stem     | Negative            | Gloss  |
|--------------|----------|---------------------|--------|
| Active verb  | u-cɨŋli: | i-cɨŋli:            | 'see'  |
| Stative verb | ma-rgí:  | i- <b>k</b> -ragi:  | 'good' |
| Noun         | ɖmári    | i- <b>k</b> -ɖamri: | 'moon' |

Negation is indicated by the prefix /i-/ and lengthening of the final vowel.

Stative verbs and nouns obligatorily appear with stative marker /ka/ in the negative form.

/ka/ → [k] in the negative form

# Ø/V alternations (rhythmic syncope)

Medial vowels are deleted in “odd syllables counting from the left edge”

Prefixes shift the position of deleted vowels.

| STEM                                  | **NEGATIVE, i-(ka-) -V                 |
|---------------------------------------|----------------------------------------|
| UR / l <i>i</i> mi l <i>i</i> ma cɪ / | / i - li m <i>i</i> li m <i>a</i> cɪ / |
| 1 2 3 4 5                             | 1 2 3 4 5 6                            |
| Syncope [ l_ mi l_ ma cɪ ]            | [ i li m_ li m_ cɪ:]                   |
| SR [ lmi lmacɪ ]                      | [ i-limlimcɪ:]                         |

\*\*Negation is indicated by the prefix /i-/and lengthening of the final vowel.

Phonetically-motivated development → morphologically conditioned alternation

## Not all prefixes trigger Ø/V alternations

| <i>Affix</i> | <i>Gloss</i> | <i>Bare stem</i> | <i>Example</i> | <i>Unattested</i> |                                      |
|--------------|--------------|------------------|----------------|-------------------|--------------------------------------|
| ma-          | stative      | rkɪmɪ            | ma-rkɪmɪ       | *ma-rikmɪ         | no vowel alternations (derivational) |
| u-           | dynamic vb   | θlibi            | u-θlibi        | *u-θilbi          |                                      |
| si-          | to put on    | kpiŋi            | si-kpiŋi       | *si-kipŋi         |                                      |
| i- -V        | negation     | pliŋɪ            | i-pilŋɪ:       | *i-pliŋɪ:         | vowel alternations (inflectional)    |
| -a:          | imperative   | brubru:          | burbur-a:      | *skul-a:          |                                      |
| l-           | plural       | vlaki            | l-valki:       | *l-vlaki:         |                                      |

suffix



Syncope does not always apply to odd-numbered vowels

Historically and in the presumed URs, Maga has no consonant clusters

So, surface CC clusters must be a result of syncope

E.g. /ɖamari/ ‘moon’ → [ɖ<sub>ɾ</sub>maɾi]

Syncope does not always apply to odd-numbered vowels

Bare stem

[**l**mi**l**maɪ]

Observed SR

→

[l\_**m**i-l\_**m**ɑɪ:]

implied deleted vowels

Syncope does not always apply to odd-numbered vowels

Bare stem

[**l****m**i**l**mɑci]

Observed SR

→

[**l**<sub>-1</sub>**m**i<sub>2</sub>**l**<sub>-3</sub>**m**a<sub>4</sub>ci:<sub>5</sub>]

implied deleted vowels

Odd vowels delete

Syncope does not always apply to odd-numbered vowels

Bare stem

[**l****m**i**l**m a c i]

Observed SR

→

[**l****\_1****m**i<sub>2</sub>**l****\_3****m**a<sub>4</sub>c i<sub>5</sub>]

implied deleted vowels

Odd vowels delete

Negative + stem

[i-l i **m** l i **m** c i]

Observed SR

→

[i<sub>1</sub> l i<sub>2</sub> **m** **\_3** l i<sub>4</sub> **m** **\_5** c i<sub>6</sub>]

implied deleted vowels

Odd vowels delete

# Syncope does not always apply to odd-numbered vowels

Bare stem

[**l****m**i**l**maɪ]  
Observed SR

→

[l<sub>1</sub>**m**i<sub>2</sub>l<sub>3</sub>**m**a<sub>4</sub>ɪ<sub>5</sub>]  
implied deleted vowels

Odd vowels delete

Negative + stem

[i-l**i****m**l**m**cɪ]  
Observed SR

→

[i<sub>1</sub>l<sub>2</sub>**m**<sub>3</sub>l<sub>4</sub>**m**<sub>5</sub>cɪ<sub>6</sub>]  
implied deleted vowels

Odd vowels delete

Negative + stative prefix + stem

[i-**k**-**q**a**m**rɪ]  
Observed SR

→

[i<sub>1</sub>**k**<sub>2</sub>-**q**a<sub>3</sub>**m**<sub>4</sub>rɪ<sub>5</sub>]  
implied deleted vowels

Even vowels delete

Stem + imperative suffix

[bu**r**u**r**aɪ]  
Observed SR

→

[bu<sub>1</sub>**r**<sub>2</sub>u<sub>3</sub>ra<sub>4</sub>ɪ]  
implied deleted vowels

Even vowels delete

**Syncope is not predictable from purely phonological factors**

# Syncope does not always apply to odd-numbered vowels

Bare stem

[**l****m**i**l**m*aci*]

Observed SR

→

[**l**<sub>1</sub>**m**<sub>2</sub>**l**<sub>3</sub>**m**<sub>4</sub>*ci*<sub>5</sub>]

implied deleted vowels

Odd vowels delete

Negative + stem

[i-**l****m**i**m***ci*]

Observed SR

→

[i<sub>1</sub>l<sub>2</sub>**m**<sub>3</sub>l<sub>4</sub>**m**<sub>5</sub>*ci*<sub>6</sub>]

implied deleted vowels

Odd vowels delete

Negative + stative prefix + stem

[i-**k**-**q**a**m***r*]

Observed SR

→

[i<sub>1</sub>**k**<sub>2</sub>-**q**<sub>3</sub>**m**<sub>4</sub>*r*<sub>5</sub>]

implied deleted vowels

Even vowels delete

Stem + imperative suffix

[bu**r**u**r**a]

Observed SR

→

[bu<sub>1</sub>**r**<sub>2</sub>u<sub>3</sub>ra<sub>4</sub>]

implied deleted vowels

Even vowels delete

Will Ø~V alternations be stable across generations? If not, what can patterns of restructuring tell us about phonological learning?

# Roadmap

Background, traditional account of  
Maga vowel alternations

**Q: why is it hard to learn?**



**Comparing two generations** of  
speakers, with different levels of  
Maga proficiency



**OT formalization:** can constraint re-  
ranking capture the difference  
between two generations? How can  
this ranking help us understand  
learning biases in phonology?

# Generations of speakers being compared

## **Generation 1: “L1, Rukai-dominant”**

Taken from Hsin (2000)

Two consultants

Birth years 1925

Dominant language is Maga Rukai  
L2: Mandarin & Japanese

790 stem-negative paradigms

## **Generation 2: “heritage speakers”**

Fieldwork data, collected May 2024

Three consultants

Birth years ~1970 - 1977

Dominant language is Mandarin  
heritage speakers of Maga Rukai

100 words re-elicited from Hsin’s list



# Result: paradigm leveling across generations

In both generations, some paradigms have leveled towards one surface form, removing  $\emptyset \sim V$  alternations.

Two possible directions of leveling: level to stem vs. level to negative form (example below)

| Direction     | Expected paradigm     | Observed paradigm    | gloss     |
|---------------|-----------------------|----------------------|-----------|
| level to stem | u-gcágcə ~ i-gacgacá: | u-gcágcə ~ i-gcəgcá: | 'scratch' |



Negative allomorph has been replaced by stem allomorph

# Result: paradigm leveling across generations

In both datasets, some paradigms have leveled towards one surface form, removing  $\emptyset \sim V$  alternations.

Two possible directions of leveling: level to stem vs. level to negative form (example below)

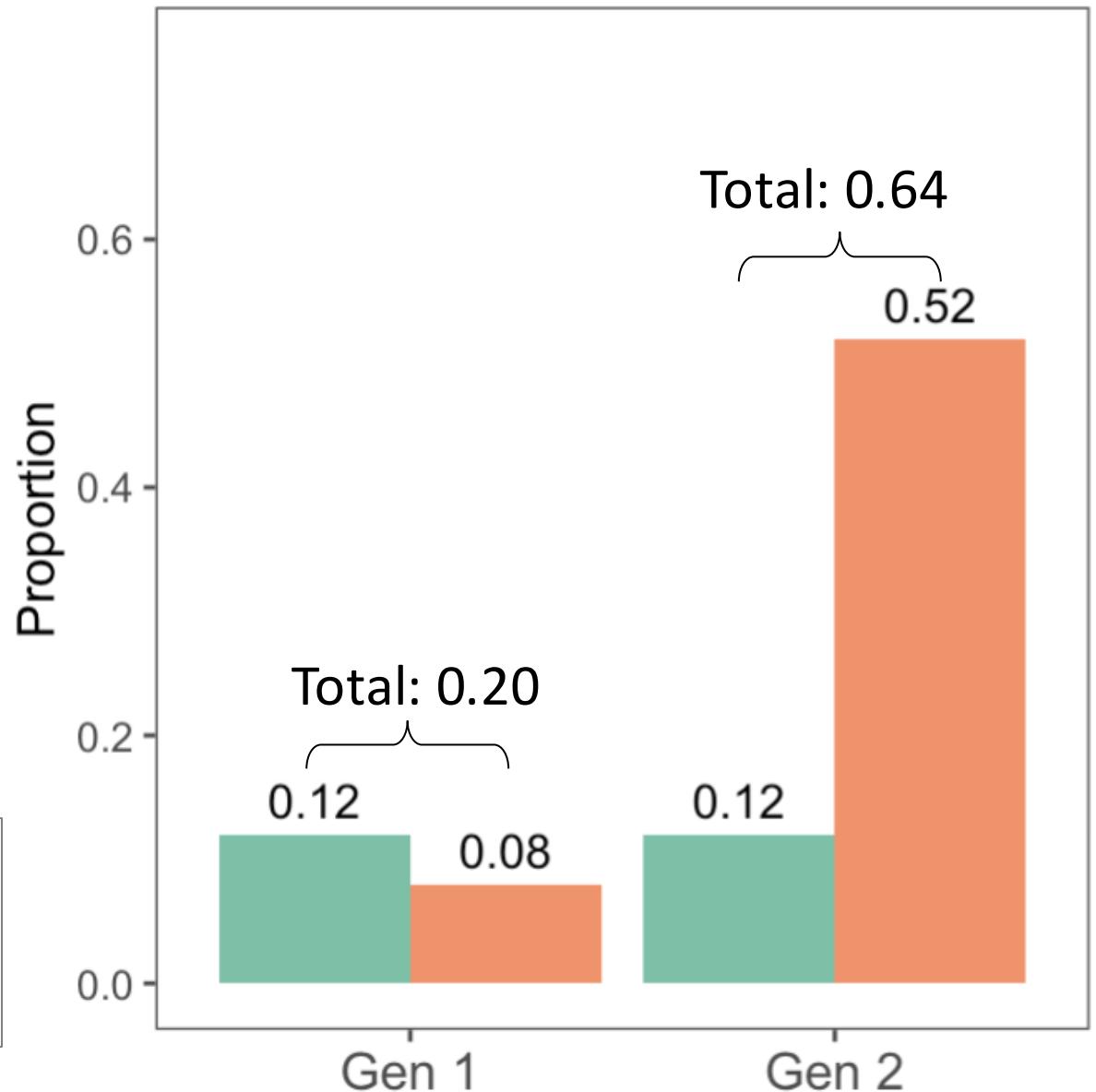
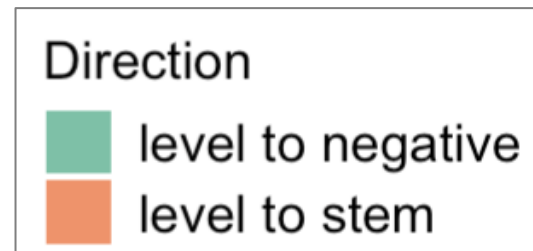
| Direction         | Expected paradigm      | Observed paradigm      | gloss     |
|-------------------|------------------------|------------------------|-----------|
| level to stem     | u-gcá'gcə ~ i-gacgacá: | u-gcá'gcə ~ i-gcə'gcá: | ‘scratch’ |
| level to negative | ma-dkiri ~ i-k-dikrî:  | ma-dikrî ~ i-k-dikrî:  | ‘heavy’   |

Stem allomorph has been replaced by negative allomorph

# Result: Rate of paradigm leveling across increases across generations

Speakers of Gen 2 have **leveled out paradigms much more extensively** than speakers of Generation 1

Growing preference for **leveling towards the stem**



# Within Gen 2, leveling varies by language use

level to negative  
level to stem

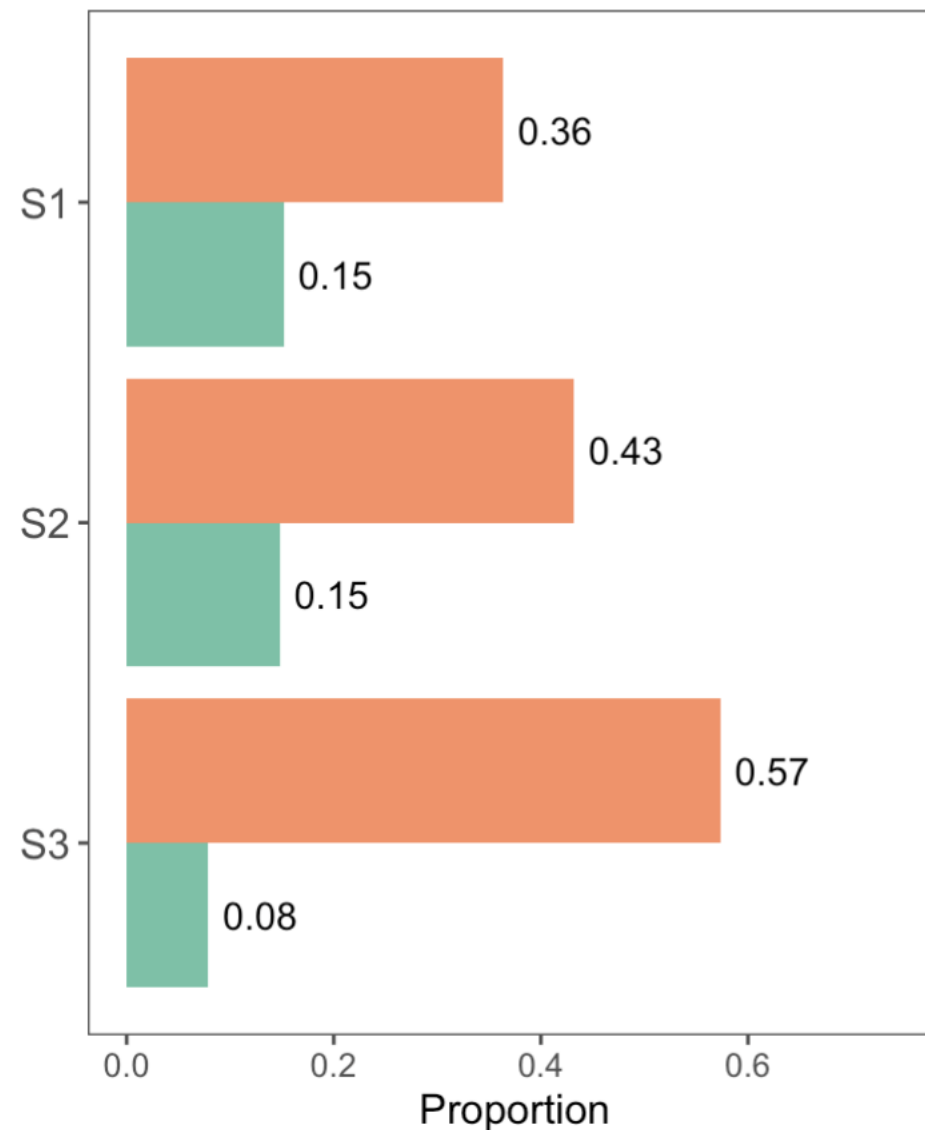
More sustained  
language use / input

**Age 61,**  
Speaks Maga  
regularly due to job

**Age 56**

**Age 54,**  
Married to speaker  
of different dialect

Less sustained  
language use / input



# Leveling interacts with general markedness restrictions

In Maga Rukai, **triconsonantal clusters are generally not allowed**, and almost never observed (\*CCC)

- OK: **sk**uɖu, **tkarv**aθi....
- BAD: **skɖ**u:, ta**krv**aθi...

However, leveling has the potential to create triconsonantal clusters:

Old paradigm  
skuɖu ~ i-k-sakɖú:

leveled to stem  
skuɖu ~ i-**k-sk**uɖú:      'stick'

**[k-] stative prefix + CC-initial root**

# Leveling interacts with general markedness restrictions

In Gen 1, leveling does **not** appear to be sensitive to the markedness restriction against CCC clusters.

Examples: CCC clusters formed by leveling

STEM                      NEGATIVE (CF. NOT LEVELED)

blavni                    i-**k-bl**avni (i-k-bulvani)

vlivla                    i-**k-vl**ivla (i-k-valvila)

Skuḍu                    i-**k-sk**uḍu (i-k-sakḍu)

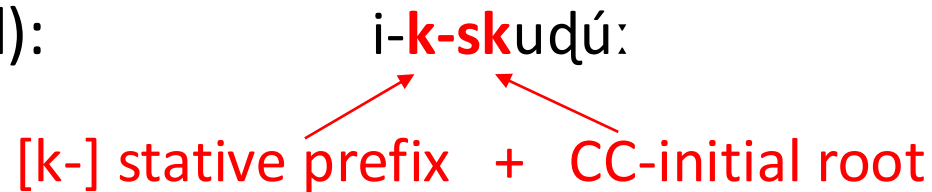
| Would leveling<br>result in a CCC<br>cluster? | p(levelled) |
|-----------------------------------------------|-------------|
| yes                                           | 0.19        |
| no                                            | 0.21        |

# Leveling interacts with general markedness restrictions

In Gen 2, speakers show **avoidance of CCC clusters**

Speakers use **[k]~[ka] allomorphy of the stative prefix**, where [ka-] is used to avoid CCC clusters.

Gen 1 (leveled):



Gen 2 (leveled + [ka-]):



# Leveling interacts with general markedness restrictions

In Gen 2, speakers show **avoidance of CCC clusters**

Speakers use **[k]~[ka] allomorphy of the stative prefix**, where [ka-] is used to avoid CCC clusters.

*Counts of [k-] vs. [ka-] in negative forms that have leveled to the stem in Gen 2*

| CC-initial | Allomorph | N  | Example                      |
|------------|-----------|----|------------------------------|
| yes        | [k-]      | 2  | [i- <b>k-b</b> lavni] (*CCC) |
|            | [ka-]     | 81 | [i-ka-vlivla]                |
| no         | [k-]      | 70 | [i-k-abunu]                  |
|            | [ka-]     | 0  | N/A                          |

} Leveling could  
create CCC cluster

} Leveling will not  
create CCC cluster



# Interim summary

## Observation

Between Gen 1 and Gen 2, there has been an **increase in paradigm leveling, removing stem alternations**

Gen 2 makes use of **novel prefix allomorphy** to remove violations of general markedness restrictions.



## Possible learning bias

Preference for **non-alternation**, Paradigm Uniformity (Benua1995; Steriade 1997, etc)

**Generality** bias (Albright 2002, Kuo 2025)

# Roadmap

Background, traditional account of Maga vowel alternations

**Q: why is it hard to learn?**



Comparing two generations of speakers, with different levels of Maga proficiency



**OT formalization:** can constraint re-ranking capture the difference between two generations? How can this ranking help us understand learning biases in phonology?

## Analysis based in Optimality Theory

- 1) Analysis  $\emptyset$ /V alternations in Gen 1
- 2) See if constraint reranking can predict outputs of Gen 2

\*Analysis can be adapted to probabilistic variants of OT, to capture more gradient patterns of leveling

# Constraints for motivating vowel alternation

Adapted from Chen (2008)

## *Constraints*

**Align-V-R:** align vowels to the right edge of a prosodic word

- Motivates syncope pattern in monomorphemes

*How it works:* assign a violation for each segment between a vowel and the right edge of the word

[t i m u s **u**]

0

# Constraints for motivating vowel alternation

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## *Constraints*

**Align-V-R:** align vowels to the right edge of a prosodic word

- Motivates syncope pattern in monomorphemes

*How it works:* assign a violation for each segment between a vowel and the right edge of the word

[t i m **u** s u]



0 + 2

# Constraints for motivating vowel alternation

Adapted from Chen (2008)

## *Constraints*

**Align-V-R:** align vowels to the right edge of a prosodic word

- Motivates syncope pattern in monomorphemes

*How it works:* assign a violation for each segment between a vowel and the right edge of the word

[t **i** m u s u]



$$0 + 2 + 4 = 6$$

# Constraints for motivating vowel alternation

Adapted from Chen (2008)

## *Constraints*

**Align-V-R:** align vowels to the right edge of a prosodic word

- Motivates syncope pattern in monomorphemes

**Align-V-L<sub>x</sub>:** align vowels to the **left** edge of a prosodic word, *in words indexed to x*.

- Indexed constraint (Pater 2007); only applies to certain morphosyntactic conditions

Motivates syncope pattern in negative form.

Alternating affixes are indexed to *x*; the resulting word (i.e. affix + base) is also indexed to *x*

In words that are indexed to *x*, vowels should instead be aligned with the **left** edge of the word

Not rhythmic syncope

Not motivated by phonetics or typology.

Requires special reference to morpheme identity

# Other constraints

## *Constraint*

## *Intuition*

**Output-Output Faith (OO-Faith):**  
violated when a morphologically  
derived form differs from the  
derivational base; really a collection of  
OO-Dep, OO-Max, and other  
faithfulness (Benua 1995)

Surface members of a paradigm should  
be similar to the citation form  
(minimizing alternation)

Motivates paradigm leveling

**\*CCC:** no triconsonantal clusters

General markedness restriction  
Phonetically motivated (Wright 1996)  
Motivates [k] – [ka] allomorphy

**Max-IO-V:** each vowel in the input (I)  
has a corresponding segment in the  
output (O)

Don't delete vowels

# Generation 1: $\emptyset$ /V alternations of stem

**Ex 1:** bare stem    /cɪŋɪli/ → [cŋɪli]

*Tableau: stem form*

| /cɪŋɪli/                  | *CCC | ALIGN-V-R | MAX-V |
|---------------------------|------|-----------|-------|
| a. cɪŋɪli (faithful)      |      | 4+2!      |       |
| → b. c_ŋɪli (deletion v1) |      | 2         | 1     |
| c. c_ŋ_li (delete v1, v2) | 1!   | 0         | 2     |



# Generation 1: $\emptyset$ /V alternations of stem

**Ex:** negation without stative prefix

**/i-ciŋɪli-V/ → [i-ciŋli:]**

*Tableau: negative form*

| /i <sub>x</sub> -ciŋɪli-V/                | *CCC | ALIGN-V-R | Max-V |
|-------------------------------------------|------|-----------|-------|
| a. [i-ciŋɪli:] <sub>x</sub> (faithful)    |      | 13        | 0     |
| b. [i-ciŋli:] <sub>x</sub> (delete v1)    |      | 8         | 1     |
| c. [i-ciŋli:] <sub>x</sub> (delete v2)    |      | 9         | 1     |
| d. [i-ciŋli:] <sub>x</sub> (delete v1,v2) | 1!   | 4         | 2     |

# Generation 1: $\emptyset$ /V alternations of stem

**Ex:** negation without stative prefix

**/i-ciŋɪli-V/ → [i-ciŋli:]**

*Tableau: negative form*

| /i <sub>x</sub> -ciŋɪli-V/               | *CCC | Align-V-<br>L <sub>x</sub> | ALIGN-V-R | Max-V |
|------------------------------------------|------|----------------------------|-----------|-------|
| a. [i-ciŋɪli:] <sub>x</sub> (faithful)   |      | 15!                        | 13        | 0     |
| b. [i-cŋɪli:] <sub>x</sub> (delete v1)   |      | 10!                        | 8         | 1     |
| → c. [i-ciŋli:] <sub>x</sub> (delete v2) |      | 9                          | 9         | 1     |
| d. [i-cŋli:] <sub>x</sub> (delete v1,v2) | 1!   | 4                          | 4         | 2     |

# Generation 1: $\emptyset$ /V alternations of stem

**Ex:** negation without stative prefix

**/i-ciŋɪli-V/ → [i-ciŋli:]**

Not important yet, but will become useful  
for modeling paradigm leveling

*Tableau: negative form*

| /i <sub>x</sub> -ciŋɪli-V/                        | *CCC | <b>Align-V-L<sub>x</sub></b> | <b>OO-Faith</b> | ALIGN-V-R | Max-V |
|---------------------------------------------------|------|------------------------------|-----------------|-----------|-------|
| a. [i-ciŋɪli:] <sub>x</sub> (faithful)            |      | 15!                          | 1               | 13        | 0     |
| b. [i-cŋɪli:] <sub>x</sub> (delete v1)            |      | 10!                          |                 | 8         | 1     |
| → c. [ <b>i-ciŋli:</b> ] <sub>x</sub> (delete v2) |      | 9                            | 1               | 9         | 1     |
| d. [i-cŋli:] <sub>x</sub> (delete v1,v2)          | 1!   | 4                            | 1               | 4         | 2     |

# Generation 1: $\emptyset$ /V alternations of stem

**Ex:** negation with stative prefix

/i-**ka**-timusu-V/ → [i-k-timsu:]

*Tableau: negative form*

| /i <sub>x</sub> -ka-timusu-V/           | *CCC | Align-V-L <sub>x</sub> | OO-Faith | ALIGN-V-R | Max-V |
|-----------------------------------------|------|------------------------|----------|-----------|-------|
| a. [i- <b>ka</b> -timsu:] <sub>x</sub>  |      | 13!                    | 1        | 15        | 1     |
| → b. [i- <b>k</b> -timsu:] <sub>x</sub> |      | 9                      | 1        | 9         | 2     |

**Intuition:** In the prefix /ka-/, vowel deletion also happens to reduce violations of **Align-V-L<sub>x</sub>**

**Constraint set captures the basic alternation pattern**

# Constraint reranking can produce all stages of change

| Stage                                            | Predicted Outputs     | Ranking                                                                         |
|--------------------------------------------------|-----------------------|---------------------------------------------------------------------------------|
| Regular alternation<br>(Gen 1)                   | [tmusu]~[i-k-timsu:]  | * <b>CCC</b> >> <b>Align-V-L<sub>x</sub></b> >><br>OO-Faith, Align-V-R >> Max-V |
| Leveling to stem<br>(Gen 1)                      | [tmusu]~[i-k-tmusu:]  | <b>OO-Faith</b> >><br>Align-V-L <sub>x</sub> >> *CCC >> Align-V-R >><br>Max-V   |
| Leveling to stem +<br>k/ka allomorphy<br>(Gen 2) | [tmusu]~[i-ka-tmusu:] | <b>OO-Faith, *CCC</b> >><br>Align-V-L <sub>x</sub> , Align-V-R >> Max-V         |

# Constraint reranking can produce all stages of change

| Stage                                                        | Predicted Outputs     | Ranking                                                                        |
|--------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------|
| Regular alternation<br>(Gen 1)                               | [tmusu]~[i-k-timsu:]  | <b>*CCC</b> >> <b>Align-V-L<sub>x</sub></b> >><br>OO-Faith, Align-V-R >> Max-V |
| Initial stage: Prioritize morphologically-driven alternation |                       |                                                                                |
| Leveling to stem<br>(Gen 1)                                  | [tmusu]~[i-k-tmusu:]  | <b>OO-Faith</b> >><br>Align-V-L <sub>x</sub> >> *CCC >> Align-V-R >><br>Max-V  |
| Leveling to stem +<br>k/ka allomorphy<br>(Gen 2)             | [tmusu]~[i-ka-tmusu:] | <b>OO-Faith, *CCC</b> >><br>Align-V-L <sub>x</sub> , Align-V-R >> Max-V        |

# Constraint reranking can produce all stages of change

| Stage                                            | Predicted Outputs     | Ranking                                                                                                                        |
|--------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Regular alternation<br>(Gen 1)                   | [tmusu]~[i-k-timsu:]  | * <b>CCC</b> >> <b>Align-V-L<sub>x</sub></b> >><br>OO-Faith, Align-V-R >> Max-V                                                |
| Leveling to stem<br>(Gen 1)                      | [tmusu]~[i-k-tmusu:]  | <b>OO-Faith</b> >><br>Align-V-L <sub>x</sub> >> *CCC >> Align-V-R >><br>Max-V<br>Intermediate : prioritize paradigm uniformity |
| Leveling to stem +<br>k/ka allomorphy<br>(Gen 2) | [tmusu]~[i-ka-tmusu:] | <b>OO-Faith</b> , * <b>CCC</b> >><br>Align-V-L <sub>x</sub> , Align-V-R >> Max-V                                               |

# Constraint reranking can produce all stages of change

| Stage                                                            | Predicted Outputs     | Ranking                                                                         |
|------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------|
| Regular alternation<br>(Gen 1)                                   | [tmusu]~[i-k-timsu:]  | * <b>CCC</b> >> <b>Align-V-L<sub>x</sub></b> >><br>OO-Faith, Align-V-R >> Max-V |
| Leveling to stem<br>(Gen 1)                                      | [tmusu]~[i-k-tmusu:]  | <b>OO-Faith</b> >><br>Align-V-L <sub>x</sub> >> *CCC >> Align-V-R >><br>Max-V   |
| Leveling to stem +<br>k/ka allomorphy<br>(Gen 2)                 | [tmusu]~[i-ka-tmusu:] | <b>OO-Faith, *CCC</b> >><br>Align-V-L <sub>x</sub> , Align-V-R >> Max-V         |
| Final : prioritize both paradigm uniformity & general markedness |                       |                                                                                 |



# Conclusion

Comparison of two generations of Maga speakers shows a shift in vowel alternations, from

complex, morphologically-driven Ø/V alternations in the stem

to:

non-alternation of stem + phonologically-motivated prefix allomorphy

**Formal analysis:** in OT, ranking OO-Faith and \*CCC above competing constraints allows us to model these patterns.

Broader finding: bias for **paradigm uniformity** and **general markedness** in contexts of generational change.

Thank you!

# Analysis with underlying /i-u-/

|         | Nouns and stative verbs                            |                                                                         | Active verbs                                         |                                                        |                                                                        |
|---------|----------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------|
| gloss   | a.<br>'moon'                                       | b.<br>'neg-stative-moon'                                                | c.<br>'act-see'                                      | d.<br>'neg-see'                                        | e.<br>'neg-active-see'                                                 |
| UR      | /ɖamari/                                           | /i-ka-ɖamari/                                                           | /u-ciŋili/                                           | /i-ciŋili-V/                                           | /i-u-ciŋili-V/                                                         |
| syncope | (ɖ <sub>-1</sub> ma <sub>2</sub> ri <sub>3</sub> ) | i-(k <sub>-1</sub> -ɖa <sub>2</sub> m <sub>-3</sub> ri <sub>4</sub> )-V | u-(c <sub>-1</sub> ŋi <sub>2</sub> li <sub>3</sub> ) | i-(c <sub>-1</sub> ŋi <sub>2</sub> li <sub>3</sub> )-V | i-( <sub>-1</sub> -ci <sub>2</sub> ŋ <sub>-2</sub> li <sub>3</sub> )-V |
| SR      | [ɖmari]                                            | [i-k-ɖamri:]                                                            | u - [cŋili]                                          | *[i-cŋili:]                                            | [i-ciŋli:]                                                             |

# (Metrical) opacity

Syncope must happen after building the metrical structure which conditions it; put another way, syncope removes the environment that conditions it

|                 |                     |                                |
|-----------------|---------------------|--------------------------------|
| UR              | /timusu/            | /tavanulu/                     |
| Metrical Struc. | (ti. <b>mu</b> ).su | (ta. <b>va</b> )( <b>u</b> lu) |
| Syncope         | tmu.su              | tvaŋ. lu                       |
| Assign stress   | tmú.su              | tvánɣ.lu                       |
| SR              | [tmúsu]             | [tvánɣlu]                      |

## *Interaction of syncope and stress assignment by syllable parity*

| Parity | UR                                                                                   | SR                                                                             | Example               |                   |                |
|--------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------|-------------------|----------------|
| Even   | /σ <sub>1</sub> σ <sub>2</sub> /                                                     | [∅ <sub>1</sub> 'σ <sub>2</sub> ]                                              | /t <b>í</b> ba/       | tb <b>á</b> :     | ‘wild yam’     |
|        | /σ <sub>1</sub> σ <sub>2</sub> <b>σ</b> <sub>3</sub> σ <sub>4</sub> /                | [∅ <sub>1</sub> 'σ <sub>2</sub> ∅ <sub>3</sub> σ <sub>4</sub> ]                | /tavan <b>u</b> lu/   | tvánɣ <b>l</b> u  | ‘chili pepper’ |
| Odd    | /σ <sub>1</sub> <b>σ</b> <sub>2</sub> σ <sub>3</sub> /                               | [∅ <sub>1</sub> 'σ <sub>2</sub> σ <sub>3</sub> ]                               | /ɖam <b>a</b> ri/     | ɖm <b>á</b> ri    | ‘moon’         |
|        | /σ <sub>1</sub> σ <sub>2</sub> σ <sub>3</sub> <b>σ</b> <sub>4</sub> σ <sub>5</sub> / | [∅ <sub>1</sub> σ <sub>2</sub> ∅ <sub>3</sub> 'σ <sub>4</sub> σ <sub>5</sub> ] | /takasul <b>u</b> ɖu/ | tkasl <b>ú</b> ɖu | ‘shrimp’       |

# Historical explanations for failure to undergo syncope are not enough

Proto-Rukai diphthongs and V?V sequences generally fail to undergo syncope in modern Rukai:

| Proto-Rukai |      | Maga   | (regular) |
|-------------|------|--------|-----------|
| *daʔanə →   | daɳi | (dɳiː) | ‘house’   |
| *payso →    | pesu | (psuː) | ‘money’   |
| *bəʔəkə →   | biki | (bkiː) | ‘pig’     |

Only accounts for **35/70 (50%)** of stem allomorphs that have a known protoform and fail to undergo syncope.

|           |       |         |                |
|-----------|-------|---------|----------------|
| *koroŋo → | kuɾɲu | (kruɲu) | ‘cloud’        |
| *dokoʔo → | duku  | (dkuː)  | ‘grow (plant)’ |
| *pəkə →   | piki  | (pkiː)  | ‘house lizard’ |